

Client Document

BioScope System User Guide

Complete instructions for account setup, sign-in, password reset, monitoring, controls (Auto and Manual), reports, alerts, devices, and settings.

Prepared for: BioScope clients and operations teams

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Purpose: This guide is designed for day-to-day users, supervisors, and administrators. It explains how to use the full web system from login to operations.

Table of Contents

1. System Overview
2. User Access and Responsibilities
3. Create an Account
4. Sign In
5. Forgot Password and Reset Password
6. Navigation and Layout
7. Dashboard
8. Sensors
9. Controls (Auto and Manual)
10. History
11. Reports and Export
12. Devices
13. Alerts
14. Settings
15. Logout and Delete Account
16. How the System Works (Data and Control Flow)
17. Troubleshooting Guide
18. Support Checklist

1) System Overview

BioScope is an environmental monitoring and control platform for smart farming and controlled environments.

- Monitors sensor readings: temperature, humidity, and light level.
- Supports remote actuator control: fan, heater, and light.
- Works with Parent Units and Child Units.

- Provides Auto mode (rule-based) and Manual mode (operator-driven).
- Provides historical records, analytics, and export files.

Component	Function
Parent Unit	Controls actuators and coordinates automation.
Child Unit	Captures sensor data and sends to platform.
Web App	Operator interface for monitoring and control.
Supabase	Stores users, readings, controls, rules, and audit logs.

2) User Access and Responsibilities

Permissions depend on account role (for example user, operator, owner, admin).

- **User:** daily monitoring and basic control tasks.
- **Operator:** operations with rule and control management.
- **Owner/Admin:** full management, including account-level and configuration operations.

Some actions may be disabled if your role does not allow them. Contact your system administrator for role updates.

3) Create an Account

1. Open the BioScope web app and click **Sign Up**.
2. Enter full name, email address, password, and confirm password.
3. Password requirement: minimum 8 characters.
4. Click **Create Account**.
5. If successful, you are redirected to the sign-in page.

Common validation messages

- Missing fields
- Password mismatch
- Password too short
- Email already exists

4) Sign In

1. Open **Login**.
2. Enter email and password.
3. Use the eye icon to show or hide password if needed.
4. Click **Sign In**.
5. On success, you are redirected to the Dashboard.

If session token expires, the app attempts automatic token refresh. If refresh fails, you are asked to sign in again.

5) Forgot Password and Reset Password

Forgot Password

1. From Login page, click **Forgot Password**.
2. Enter the registered email address.
3. Click **Send Reset Link**.
4. Check email inbox (and spam folder).

Reset Password

1. Open the reset link from email.
2. Enter new password and confirm password.
3. Password requirement: minimum 8 characters.
4. Submit and sign in with the new password.

If link is invalid or expired, request another reset link from the Forgot Password page.

6) Navigation and Layout

After login, use the left navigation menu:

Section	Items	Purpose
Monitor	Dashboard, Sensors, Controls	Live operations and control.
Data	History, Reports	Historical analysis and exports.
Manage	Devices, Alerts, Settings	Configuration and administration.

Top bar includes **Refresh** and latest reading timestamp.

7) Dashboard

Dashboard provides a quick operational overview for each Parent Unit.

- Live cards for temperature, humidity, and light.
- Online/offline status per device.
- Mode switch (Auto or Manual) at device level.
- Slot summary showing assigned actuator and ON/OFF status.
- Active alerts panel.
- Quick statistics (total devices, active alerts, parent/child counts).

Click **Manage** in slot area to jump directly to Controls page for detailed actuator operations.

8) Sensors

- Select a Child Unit from device tabs.
- View live metric cards for temperature, humidity, and light.

- View charts for last 1 hour or 24 hours.
- View latest reading details with timestamp.

Status interpretation

- Temperature high if above threshold.
- Humidity high/low if outside configured range.
- Light shown in lux.

9) Controls (Auto and Manual)

Controls page is the core actuator control interface.

9.1 Mode Control

- **Manual mode:** operator toggles actuator state directly.
- **Auto mode:** system evaluates active rules and applies states automatically.
- When switching from Manual to Auto, the system evaluates immediately using latest reading.

9.2 Actuator Slots

- Two physical actuator slots are supported.
- Each slot can be assigned one logical actuator: **fan**, **heater**, or **light**.
- The same logical actuator cannot be assigned to both slots.

9.3 Rule Behavior

- Automation decision is shown for each slot.
- If no active rule matches, previous state may be preserved or shown as no match based on current context.
- Fan and Heater are mutually exclusive.

9.4 Practical workflow

1. Assign slot devices correctly.
2. Ensure required rules are active.
3. Switch to Auto mode.
4. Verify automation decision and slot ON/OFF indicators.

If a device is not assigned to any slot, it will not be controlled in slot operations.

10) History

- Select a Child Unit.
- Search records (temperature, humidity, reading id).
- Sort by newest or oldest.
- Browse paginated readings table.
- Export filtered result to CSV or JSON.

History is ideal for audits, inspections, and trend checks over time.

11) Reports and Export

Reports page supports device analytics and document generation.

- Per-device statistics: average, minimum, maximum.
- Export window selection: 1 hour, 24 hours, 7 days.
- Export formats: CSV, JSON, PDF.
- Generate full consolidated PDF report across devices.

Format	Best use
CSV	Spreadsheet analysis and external calculations.
JSON	System integrations and technical processing.
PDF	Management sharing, formal reporting, and client communication.

12) Devices

12.1 Register Device

1. Open Devices page and click **Register Device**.
2. Choose device type: Parent Unit or Child Unit.
3. Enter device id, name, and location.
4. For Child Unit, select parent device.
5. Save registration.

12.2 Device actions

- View status and hardware metadata.
- Ping and restart actions (operational tools).
- Remove device (if authorized).

13) Alerts

13.1 Live Alerts

- See current critical/warning alerts.
- Acknowledge alerts after review.

13.2 Alert Rules

- Enable or disable rules.
- Review condition, severity, and channel per rule.

13.3 Notification Channels

- Email and in-app channels are visible in channel status panel.
- SMS availability depends on deployment configuration.

14) Settings

Settings provides user-level behavior and account preferences.

General

- Theme (light or dark)
- Auto refresh toggle and interval
- Temperature unit
- Language

Notifications

- Email alerts
- Push alerts
- SMS alerts

Data and Privacy

- Data retention preference
- Export all data
- Delete account (with confirmation)

Account

- View profile summary and role badge
- Logout

15) Logout and Delete Account

Logout

1. Open Settings.
2. Click **Logout**.

Delete Account

1. Open Settings.
2. Click **Delete** under Data and Privacy.
3. Confirm deletion in modal dialog.

Account deletion is permanent and cannot be undone.

16) How the System Works (Data and Control Flow)

Data flow

1. Child unit captures sensor values.
2. Readings are sent to backend and stored in database.
3. Dashboard/Sensors/History/Reports fetch these records and display them.

Manual control flow

1. User toggles actuator in Controls.
2. Control action is recorded.
3. Actuator status is updated and reflected in UI.

Auto control flow

1. User switches mode to Auto.
2. System evaluates active automation rules against latest reading.
3. Resolved state is applied to fan, heater, and light.
4. Slot ON/OFF indicators and decision text are updated.

17) Troubleshooting Guide

Issue	Possible Cause	Action
Cannot sign in	Wrong credentials or expired session	Re-enter credentials, use Forgot Password, then try again.
No new sensor data	Device offline or timestamp mismatch	Check device power/network and use Refresh in top bar.
Auto mode not changing actuator state	No active matching rule, wrong slot assignment, or no latest reading	Verify rules are active, slot assignment is correct, and latest reading exists.
Rule decision says ON but actuator appears OFF	State sync delay	Wait one refresh cycle or click Refresh; verify mode and slot mapping.
Password reset email not received	Email typo, inbox filtering, or rate limiting	Check email address, spam folder, then retry after a short wait.

18) Support Checklist (Send with support ticket)

- User email and role
- Page where issue occurred
- Device id (for example P1, C1)
- Timestamp of issue
- Screenshot of screen and any error message
- Whether mode was Auto or Manual

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This document is intended for operational use by client teams. Keep it with your standard operating procedures.